



replicate result



replicate result

heights and
approval ratings
of presidents



	height	approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9



	height	approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?



height



approval
rating

Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?
2. Pearson's r ?



	height	approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?
2. Pearson's r ?
3. Regression equation and regression line?



	height	approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?
2. Pearson's r ?
3. Regression equation and regression line?
4. Size r^2 ?

	 height	 approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?

	 height	 approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?

Is there a linear relationship between the two variables?



	height	approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?

Is there a linear relationship between the two variables?



scatterplot

	 height	 approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?

Is there a linear relationship between the two variables?



scatterplot



dependent variable and
independent variable



	height	approval rating
Obama	185	47.0
Bush jr	182	49.9
Clinton	188	55.1
Bush sr	188	60.9

1. Linear relationship?

Is there a linear relationship between the two variables?

scatterplot

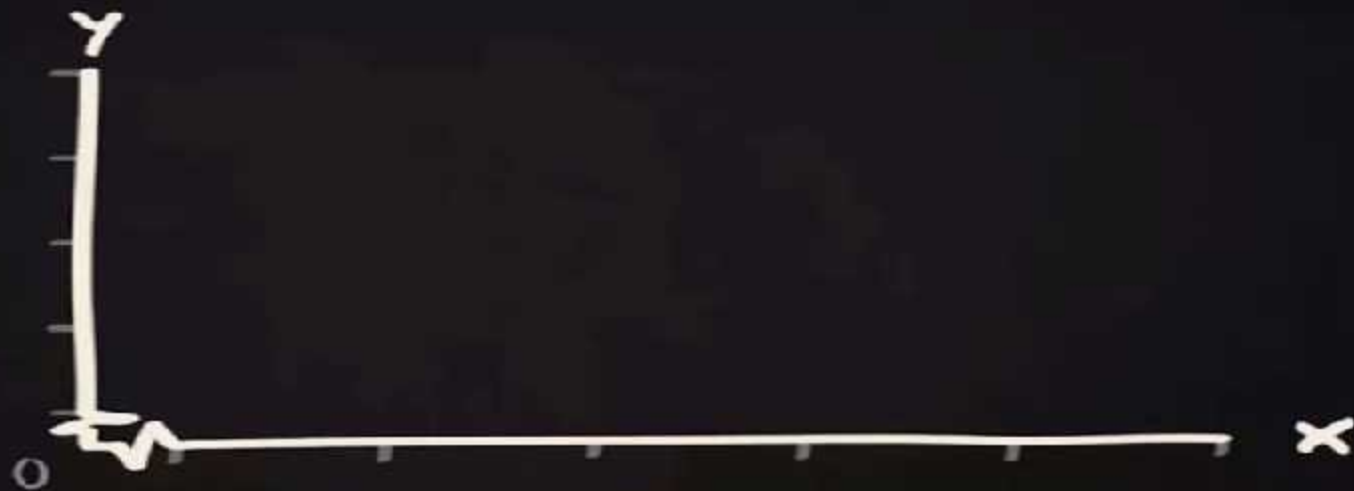
dependent variable and
independent variable

1. Linear relationship?



1. Linear relationship?


DV
approval
rating




IV
height

1. Linear relationship?



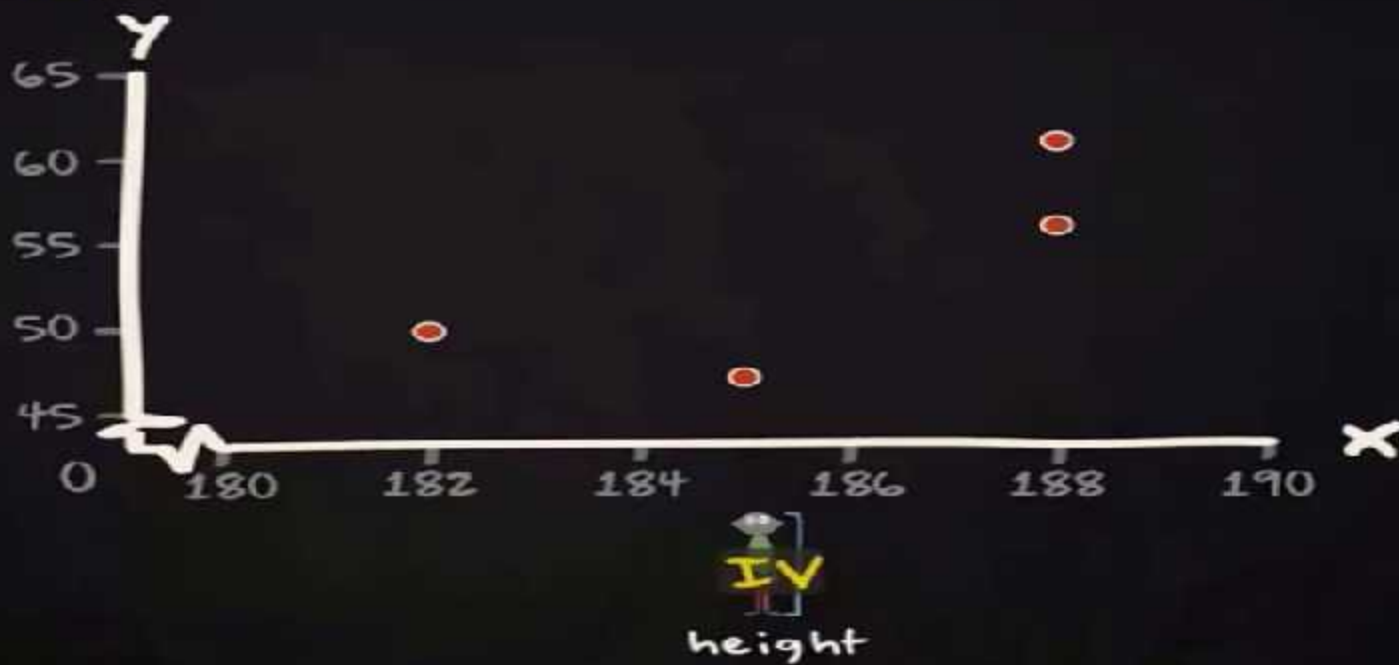
1. Linear relationship?



1. Linear relationship?

↓
YES

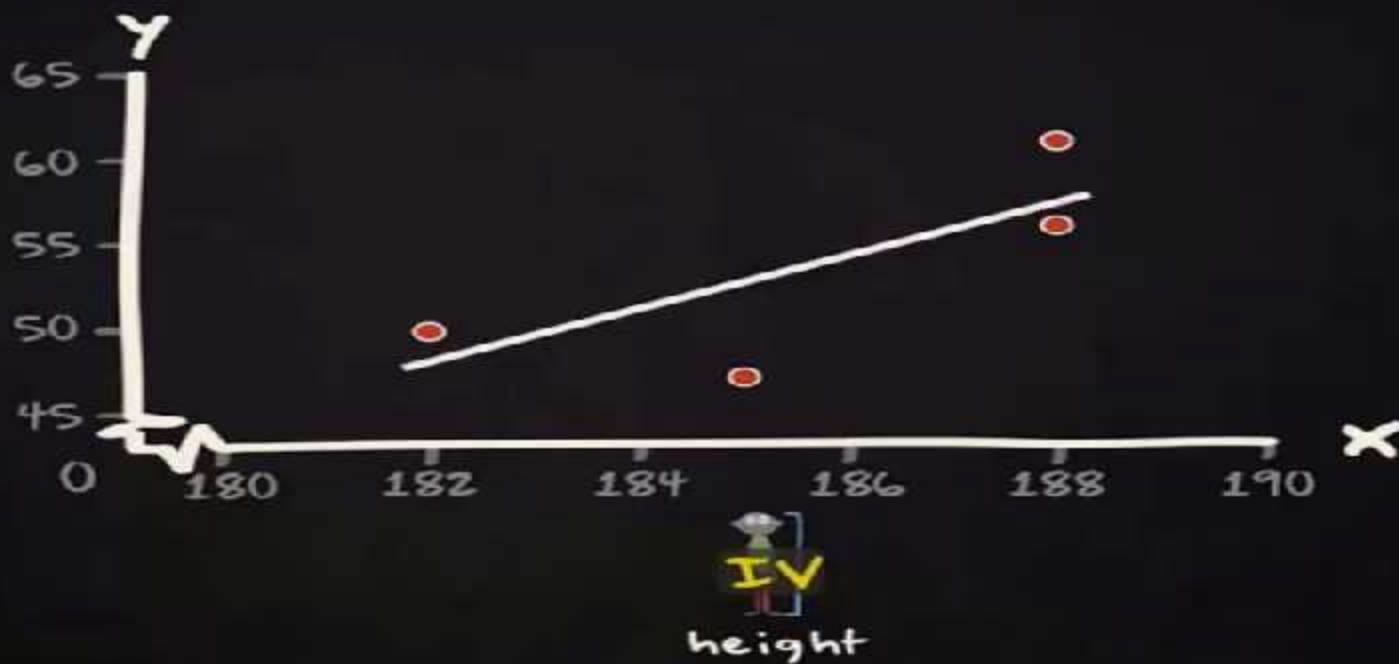

DV
approval
rating



1. Linear relationship?

↓
YES


DV
approval
rating



1. Linear relationship?

↓
YES


DV
approval
rating



2. Pearson's r ?

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$$r = \frac{\sum Z_x Z_y}{n - 1}$$

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$\bar{x}$ $\bar{y}$
 s_x s_y

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y}$$

$$s_x = 2.87 \quad s_y$$

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$S_x = 2.87 \quad S_y = 6.12$$



	x	Z _x	y		
Obama	185		47.0		
Bush jr	182		49.9		
Clinton	188		55.1		
Bush sr	188		60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y		
Obama	185		47.0		
Bush jr	182		49.9		
Clinton	188		55.1		
Bush sr	188		60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y		
Obama	185	$\frac{185 - 185.75}{2.87}$	47.0		
Bush jr	182		49.9		
Clinton	188		55.1		
Bush sr	188		60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y		
Obama	185	-0.26	47.0		
Bush jr	182		49.9		
Clinton	188		55.1		
Bush sr	188		60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y		
Obama	185	-0.26	47.0		
Bush jr	182	-1.31	49.9		
Clinton	188	0.78	55.1		
Bush sr	188	0.78	60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	
Obama	185	-0.26	47.0		
Bush jr	182	-1.31	49.9		
Clinton	188	0.78	55.1		
Bush sr	188	0.78	60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	
Obama	185	-0.26	47.0	$\frac{47 - 53.23}{6.12}$	
Bush jr	182	-1.31	49.9		
Clinton	188	0.78	55.1		
Bush sr	188	0.78	60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	
Obama	185	-0.26	47.0	-1.02	
Bush jr	182	-1.31	49.9		
Clinton	188	0.78	55.1		
Bush sr	188	0.78	60.9		

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	
Obama	185	-0.26	47.0	-1.02	
Bush jr	182	-1.31	49.9	-0.54	
Clinton	188	0.78	55.1	0.31	
Bush sr	188	0.78	60.9	1.26	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y
Obama	185	-0.26	47.0	-1.02	
Bush jr	182	-1.31	49.9	-0.54	
Clinton	188	0.78	55.1	0.31	
Bush sr	188	0.78	60.9	1.26	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y
Obama	185	-0.26	47.0	-1.02	-0.26 * -1.02
Bush jr	182	-1.31	49.9	-0.54	
Clinton	188	0.78	55.1	0.31	
Bush sr	188	0.78	60.9	1.26	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y
Obama	185	-0.26	47.0	-1.02	0.27
Bush jr	182	-1.31	49.9	-0.54	
Clinton	188	0.78	55.1	0.31	
Bush sr	188	0.78	60.9	1.26	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{X} = 185.75 \quad \bar{Y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y
Obama	185	-0.26	47.0	-1.02	0.27
Bush jr	182	-1.31	49.9	-0.54	0.71
Clinton	188	0.78	55.1	0.31	0.24
Bush sr	188	0.78	60.9	1.26	0.98

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Finished!

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y
Obama	185	-0.26	47.0	-1.02	0.27
Bush jr	182	-1.31	49.9	-0.54	0.71
Clinton	188	0.78	55.1	0.31	0.24
Bush sr	188	0.78	60.9	1.26	0.98

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y
Obama	185	-0.26	47.0	-1.02	0.27
Bush jr	182	-1.31	49.9	-0.54	0.71
Clinton	188	0.78	55.1	0.31	0.24
Bush sr	188	0.78	60.9	1.26	0.98

} Σ

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$r = \frac{2.203}{4}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y	
Obama	185	-0.26	47.0	-1.02	0.27	} Σ
Bush jr	182	-1.31	49.9	-0.54	0.71	
Clinton	188	0.78	55.1	0.31	0.24	
Bush sr	188	0.78	60.9	1.26	0.98	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1} \rightarrow r = \underline{2.203}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y	} Σ
Obama	185	-0.26	47.0	-1.02	0.27	
Bush jr	182	-1.31	49.9	-0.54	0.71	
Clinton	188	0.78	55.1	0.31	0.24	
Bush sr	188	0.78	60.9	1.26	0.98	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1} \rightarrow r = \frac{2.203}{4 - 1}$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x * Z _y	} Σ
Obama	185	-0.26	47.0	-1.02	0.27	
Bush jr	182	-1.31	49.9	-0.54	0.71	
Clinton	188	0.78	55.1	0.31	0.24	
Bush sr	188	0.78	60.9	1.26	0.98	

2. Pearson's r?

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\rightarrow r = \frac{2.203}{4 - 1} = 0.73$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x *Z _y
Obama	185	-0.26	47.0	-1.02	0.27
Bush jr	182	-1.31	49.9	-0.54	0.71
Clinton	188	0.78	55.1	0.31	0.24
Bush sr	188	0.78	60.9	1.26	0.98

$$\left. \vphantom{\begin{matrix} 0.27 \\ 0.71 \\ 0.24 \\ 0.98 \end{matrix}} \right\} \Sigma = 2.203$$

2. Pearson's r?

↓
0.73

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

$$\rightarrow r = \frac{2.203}{4 - 1} = 0.73$$

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$Z = \frac{x - \bar{x}}{s}$$

	x	Z _x	y	Z _y	Z _x *Z _y
Obama	185	-0.26	47.0	-1.02	0.27
Bush jr	182	-1.31	49.9	-0.54	0.71
Clinton	188	0.78	55.1	0.31	0.24
Bush sr	188	0.78	60.9	1.26	0.98

$$\left. \vphantom{\begin{array}{c} 0.27 \\ 0.71 \\ 0.24 \\ 0.98 \end{array}} \right\} \Sigma = 2.203$$

3. Regression equation and regression line?

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$$b = r \left(\frac{s_y}{s_x} \right)$$

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$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

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$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\bar{x} = 185.75 \quad \bar{y} = 53.23$$

$$s_x = 2.87 \quad s_y = 6.12$$

$$r = 0.73$$

3. Regression equation and regression line?

$0.73 \cdot (6.12 / 2.87) = 1.56$ ← $b = r \left(\frac{s_y}{s_x} \right)$

$a = \bar{y} - b(\bar{x})$

regression line

$\bar{x} = 185.75$ $\bar{y} = 53.23$
 $s_x = 2.87$ $s_y = 6.12$
 $r = 0.73$

```
graph LR; A["0.73 * (6.12 / 2.87) = 1.56"] --> B["b = r (sy/sx)"]; B --> C["a = y-bar - b(x-bar)"]; C --> D["regression line"]; E["x-bar = 185.75, y-bar = 53.23, s_x = 2.87, s_y = 6.12, r = 0.73"] --> A;
```

3. Regression equation and regression line?

$0.73 \cdot (6.12 / 2.87) = 1.56$ ← $b = r \left(\frac{s_y}{s_x} \right)$

$53.23 - 1.56(185.75) = -237.11$ ← $a = \bar{y} - b(\bar{x})$

$\bar{x} = 185.75$ $\bar{y} = 53.23$
 $s_x = 2.87$ $s_y = 6.12$
 $r = 0.73$

regression line

```
graph LR; A["0.73 * (6.12 / 2.87) = 1.56"] --> B["b = r * (sy / sx)"]; B --> C["regression line"]; D["53.23 - 1.56 * 185.75 = -237.11"] --> E["a = y-bar - b * x-bar"]; E --> C; F["x-bar = 185.75, y-bar = 53.23, s_x = 2.87, s_y = 6.12, r = 0.73"] --> A; F --> D;
```

3. Regression equation and regression line?

$$0.73 * (6.12 / 2.87) = 1.56$$

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$53.23 - (1.56 * 185.75) = -237.11$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

3. Regression equation and regression line?

$$0.73 * (6.12 / 2.87) = 1.56$$

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$53.23 - (1.56 * 185.75) = -237.11$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11$$

3. Regression equation and regression line?

$$0.73 * (6.12 / 2.87) = 1.56$$

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$53.23 - (1.56 * 185.75) = -237.11$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

intercept

3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

intercept

mathematical purposes

3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 182$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 182$$

$$\hat{y} = -237.11 + 1.56 \cdot 182$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 182$$

$$\hat{y} = -237.11 + 1.56 \cdot 182$$

$$\hat{y} = 46.81$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 182$$

$$\hat{y} = -237.11 + 1.56 \cdot 182$$

$$\hat{y} = 46.81$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 188$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 188$$

$$\hat{y} = -237.11 + 1.56 \cdot 188$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 188$$

$$\hat{y} = -237.11 + 1.56 \cdot 188$$

$$\hat{y} = 56.17$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

$$x = 188$$

$$\hat{y} = -237.11 + 1.56 \cdot 188$$

$$\hat{y} = 56.17$$



3. Regression equation and regression line?

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression line

$$\hat{y} = a + bx$$

$$\hat{y} = -237.11 + 1.56x$$



4. Size r^2 ?

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$$r = 0.73$$

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$$r = 0.73$$

$$r^2 = 0.73 * 0.73$$

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$$r = 0.73$$

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$$r^2 = 0.53$$

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$$r = 0.73$$

$$r^2 = 0.73 * 0.73$$

$$r^2 = 0.53$$



<
53 %



3. Size r^2 ?

$$r = 0.73$$

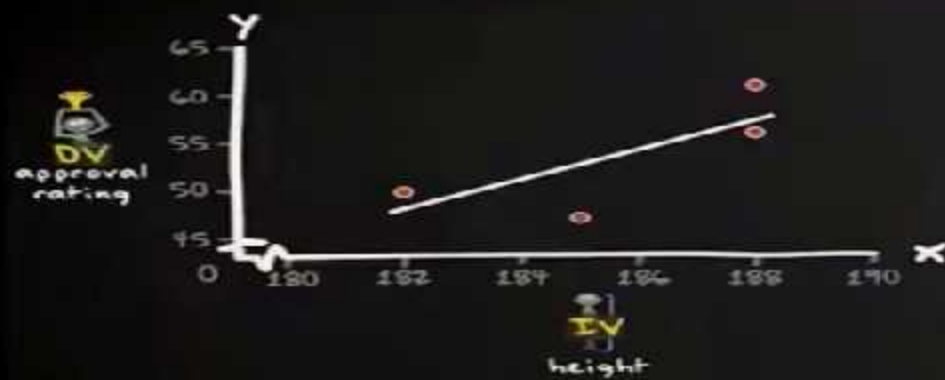
$$r^2 = 0.73 * 0.73$$

$$r^2 = 0.53$$

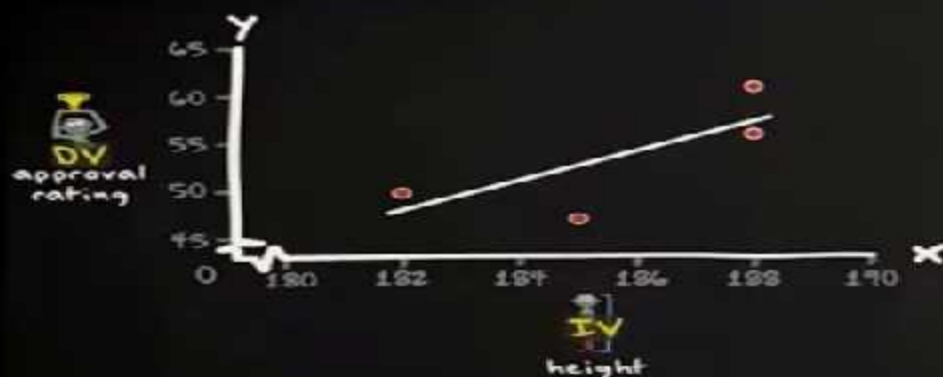


53 % of the variation
in y is explained by x

summary



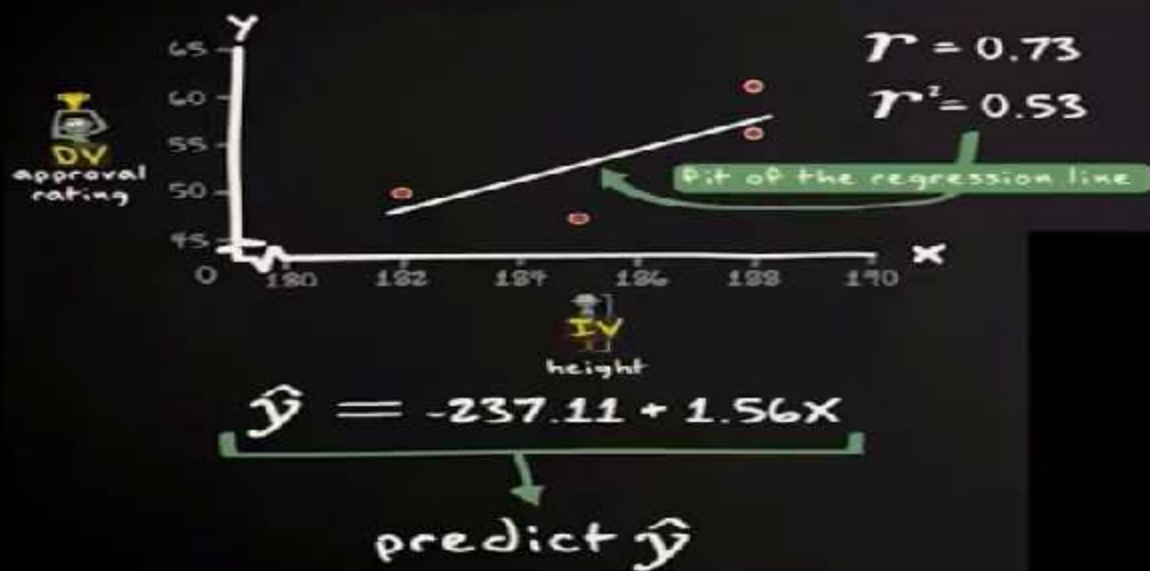
summary



$$\hat{y} = -237.11 + 1.56x$$

predict $\hat{y}$

summary





taller is better